MARY SYAMALA

GENAI ENGINEER | AI/ML ENGINEER | NLP | PYTHON | DATA ENGINEERING | CLOUD
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PROFESSIONAL SUMMARY

GenAI Engineer with senior-level expertise delivering AI/ML and NLP solutions. Currently leading the development of a real-time market-intelligence platform that ingests live news, applies FinBERT sentiment analysis, and serves evidence-grounded GenAI workflows via FastAPI. Proven track record in building scalable data pipelines, model governance, and translating stakeholder needs into reliable AI products. Aiming to leverage generative AI and data-engineering skills to drive innovative, data-driven outcomes for the organization.

TECHNICAL SKILLS

Generative AI, LLMs & NLP	: LangChain, OpenAI API, LLaMA-2-7B-Chat, FinBERT, prompt engineering, evidence-grounded generation, citations, guardrails, embeddings, vector search, sentiment analysis
Machine Learning & Model Evaluation	: Scikit-learn, LightGBM, deep learning, supervised learning, classification, time-series modeling, feature engineering, model evaluation, walk-forward validation, out-of-sample testing
Programming, APIs & Data Engineering	: Python, SQL, PySpark, Java, FastAPI, REST APIs, WebSockets, Parquet, ETL/ELT, data transformation, dbt, API/XML integration
Platforms, Vector Search & Databases	: PostgreSQL, pgvector, Snowflake, Databricks, BigQuery, SQL Server, MongoDB, Oracle
Cloud, MLOps & DevOps	: AWS S3/Athena, Microsoft Azure data services, GCP, Apache Airflow, Terraform, GitHub Actions, CI/CD, Git, pipeline monitoring, automated testing
Analytics, Data Quality & Reporting	: Power BI, Tableau, Excel, Google Analytics, KPI dashboards, data profiling, reconciliation, data validation, data quality, anomaly detection

PROFESSIONAL EXPERIENCE

Symplore Inc. **Sep 2026 – Present**
GenAI Engineer *Michigan*

- Developing a real-time U.S. market intelligence platform integrating market/news data, FinBERT NLP sentiment, event-reaction analytics, FastAPI services, and GenAI/RAG workflows for equity research and monitoring.
- Engineering Python/FastAPI REST and WebSocket pipelines for ingestion, validation, deduplication, entity/ticker mapping, and Parquet processing, delivering AI-ready data through PostgreSQL and pgvector serving layers.
- Processing a 2015–2026 archive of ~95K market-news articles, implementing deduplication, relevance/entity mapping, and FinBERT sentiment scoring across 61K+ articles for NLP, analytics, and ML evaluation.
- Building point-in-time data pipelines for prices, trading calendars, corporate events, and news with lineage, leakage controls, quarantine rules, reproducible runs, and idempotent processing to ensure reliable model training and evaluation.
- Supporting ML and GenAI development using Logistic Regression, LightGBM, walk-forward/time-based validation, bootstrap confidence intervals, and out-of-sample testing, while building evidence-grounded RAG/LLM workflows with citations, numeric validation, prompt-injection defenses, and hallucination safeguards.
- Supporting production deployment across AWS ECS Fargate, S3, RDS PostgreSQL/pgvector, Cognito, Secrets Manager, and CloudWatch, with GitHub Actions CI/CD and OIDC, automated testing, linting, regression checks, and API/data-quality validation.

InCom Technologies **Feb 2025 - Aug 2026**
AI Data Engineer *Remote*

- Built and maintained ETL/ELT pipelines to ingest, transform, and validate structured and semi-structured data for AI/ML models, analytics, and data-driven applications**.**
- Developed automated Python, SQL, and API-based data pipelines to prepare AI-ready datasets, reducing processing time by 25% and improving data availability and reliability.
- Implemented data validation, reconciliation, and quality checks to detect missing, inconsistent, and anomalous records, improving the reliability of datasets used for model training and inference**.**
- Collaborated with AI/ML and business teams to translate requirements into reusable datasets, feature pipelines, automated workflows, and model-ready data**,** while supporting Power BI and Tableau reporting.

Advanced Knowledge Tech **Oct 2023 - Jan 2025**
Data Analyst *Texas*

- Engineered and automated ETL, data validation, and audit workflows using Python, SQL, and APIs, reducing manual reporting effort by 40% and improving pipeline reliability and repeatability.
- Implemented data profiling and quality-control frameworks to identify anomalies, missing values, duplicates, and inconsistent records, improving the accuracy and reliability of downstream analytical datasets.
- Analyzed large-scale operational and financial datasets using SQL and advanced Excel, identifying process inefficiencies and data-driven optimization opportunities that contributed to a 10% cost reduction.

- Designed interactive Power BI dashboards and KPI monitoring solutions for operational and vendor performance, increasing visibility and reducing issue-detection time by 30%.

DXC Technology **Mar 2021 - May 2022**

Junior Data Analyst *India*

- Processed, cleansed, and analyzed 100K+ records daily using SQL, Python, and Excel, delivering accurate, analytics-ready datasets for operational, vendor, and financial reporting.
- Built and validated ETL pipelines with automated data-quality checks, improving data accuracy, consistency, and reliability across downstream analytics and reporting workflows.
- Conducted trend, variance, and cost analysis across operational and financial datasets, identifying inefficiencies and translating findings into actionable recommendations for finance and operations teams.
- Developed interactive Power BI and Tableau dashboards, partnering with stakeholders to streamline recurring reporting, strengthen KPI visibility, and improve data-quality and analytics processes

PROJECTS

UTD Generative AI Chatbot | *UT Dallas*

- Built a university-focused conversational AI application using LangChain, OpenAI API, web scraping, and LLaMA-2-7B-Chat; use prompt engineering and domain content to improve response relevance and reliability.

Stock Market Prediction Using News Sentiment | *JNTUH*

- Developed a deep-learning and time-series project using news sentiment signals to explore stock-market movement prediction, combining data preparation, model experimentation, and predictive analysis.

EDUCATION

The University of Texas at Dallas
M.S., Business Analytics & Artificial Intelligence

Jawaharlal Nehru Technological University, Hyderabad
B.Tech, Computer Science & Engineering

CERTIFICATIONS

- Google Project Management
- Google Data Analytics
- Microsoft Azure Data Fundamentals
- Supervised Learning in R: Classification (DataCamp)
- Exploratory Data Analysis in R (DataCamp)

LEADERSHIP & ENTREPRENEURSHIP

myclicks_c Photography (100+ clients) | *Founder*

Photography Club & Sports Club, BRECW | *President*

Under-19 Women's Cricket Team, Hyderabad Cricket Association | *Captain*